

Intrinsic Timing and Position Reconstruction with Adjacent SiPMs

Fourth- and Twentieth-Detected-Hit Timing, Local Position Reconstruction and EndTop Context

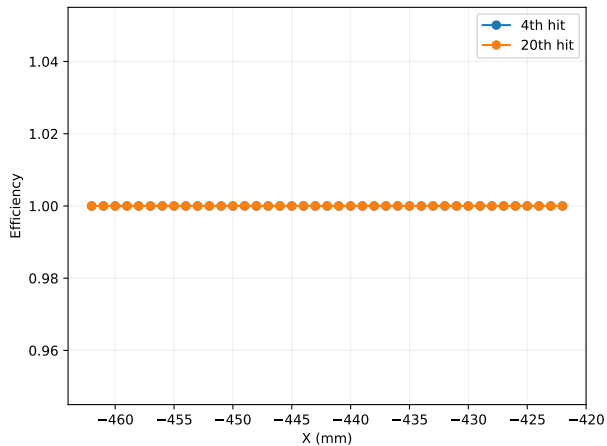
René Ríos Torres

Universidad de La Serena

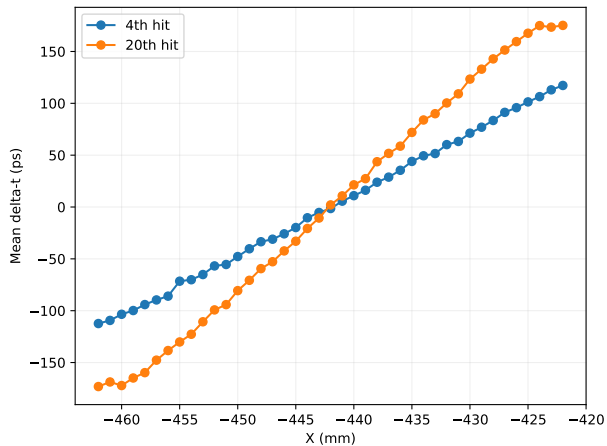
June 12, 2026

- EXEC 11 4PE reproduction gate: **PASS**.
- Native and matched samples coincide: 4PE and 20PE efficiencies are 100%.
- 20th-hit differential width: 99.8 ps, versus 78.5 ps for 4th hit.
- Cross-validated X RMS68: 10.32 mm at 20PE, versus 13.34 mm at 4PE.
- Higher thresholds improve X RMS68 here, while calibration nonlinearity rises.
- Intrinsic simulation prediction; 20th detected hit is not waveform 20% CFD.

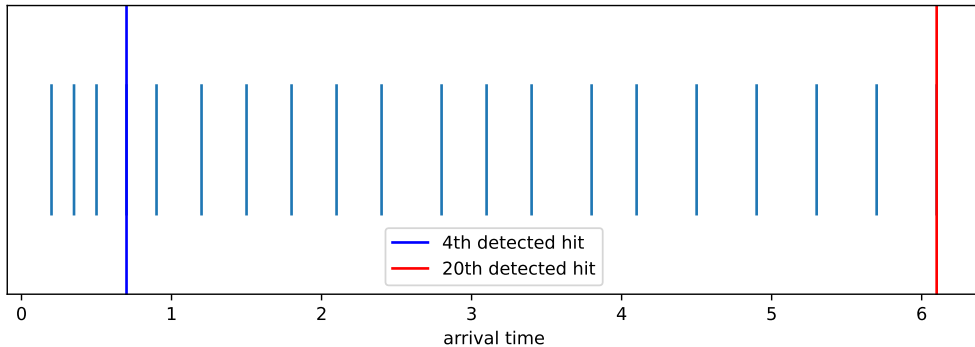
Detector and fine scan



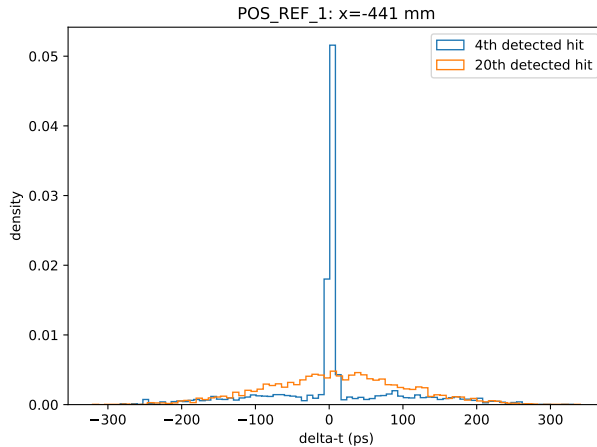
Available datasets



From hits to observables



Ordered photon arrivals

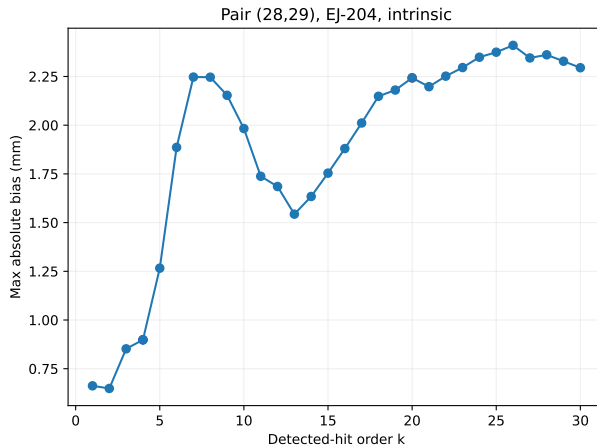


Critical distinction: 20th hit is not 20 percent CFD

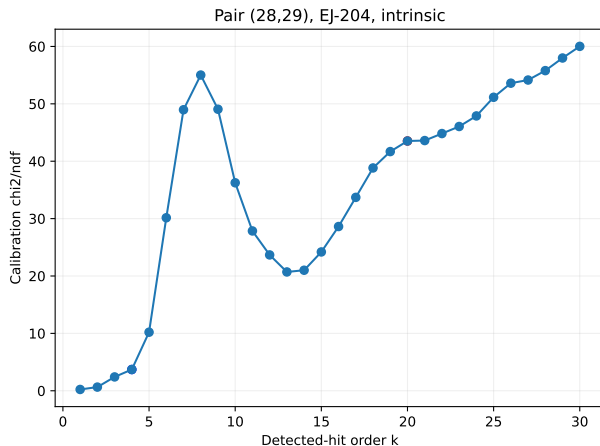
20th detected-hit timing $\neq$ 20% constant-fraction timing.

Order statistic versus waveform threshold; one pair versus 64-channel combination.

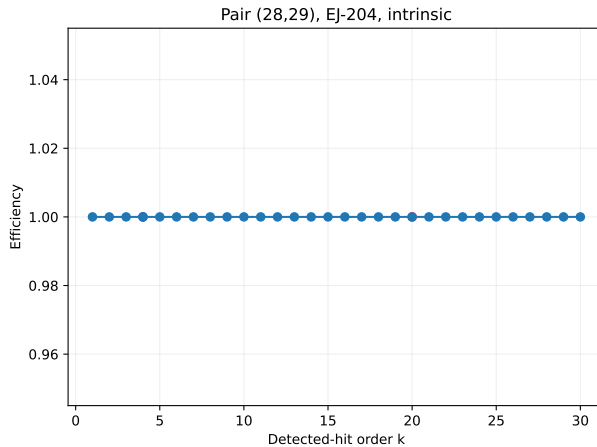
Common and differential timing



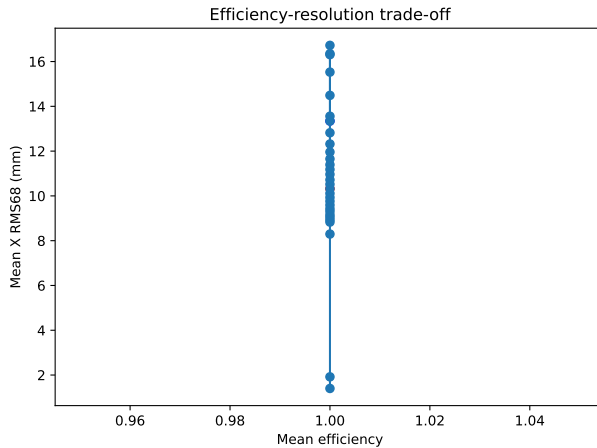
Native versus matched samples



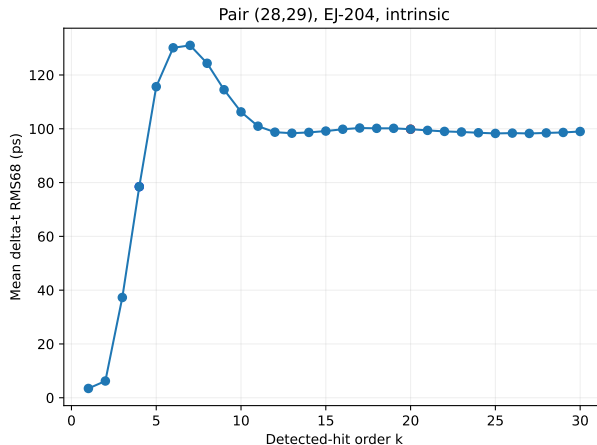
4PE reproduction gate



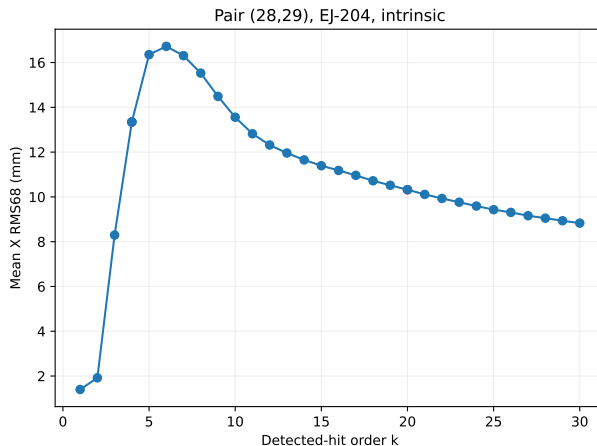
Raw temporal response



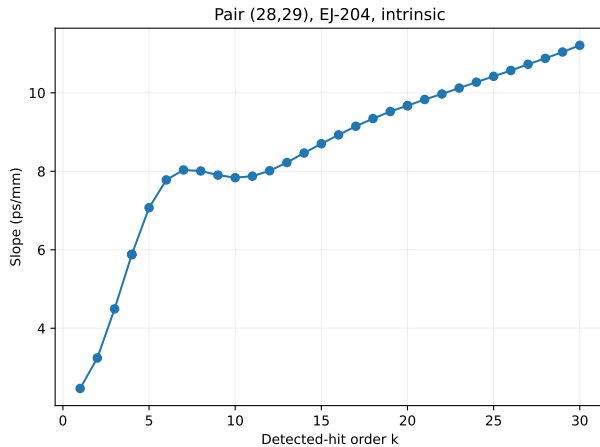
Efficiency versus X



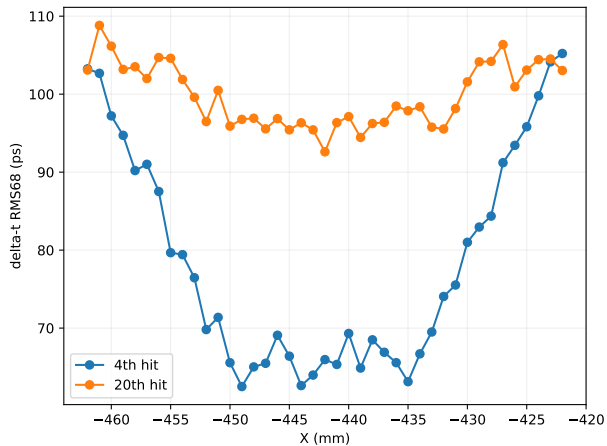
Mean differential time



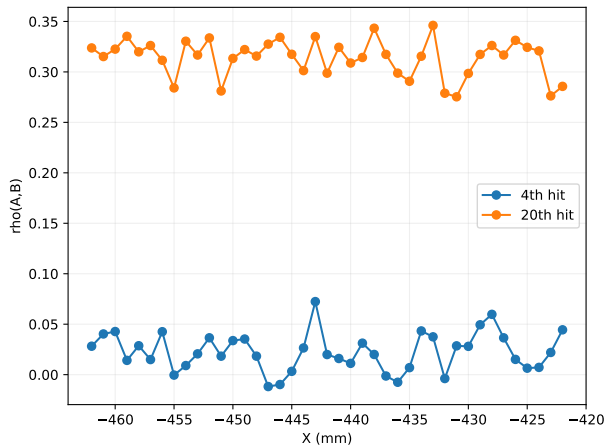
Timing width versus X



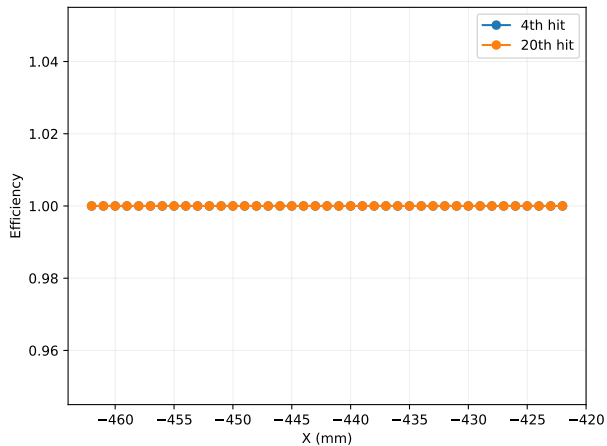
A-B correlation



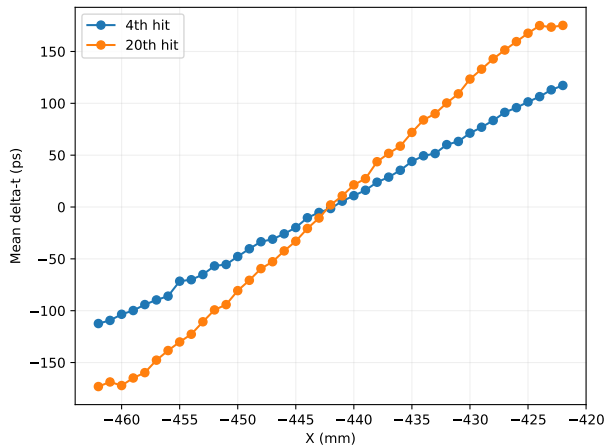
4PE calibration



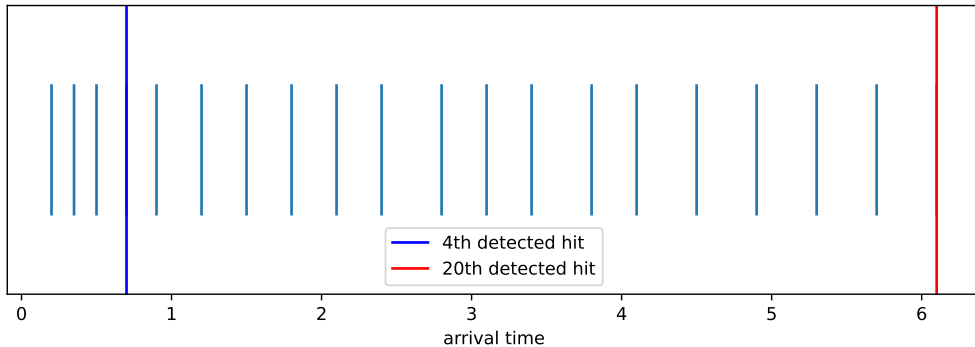
20PE calibration



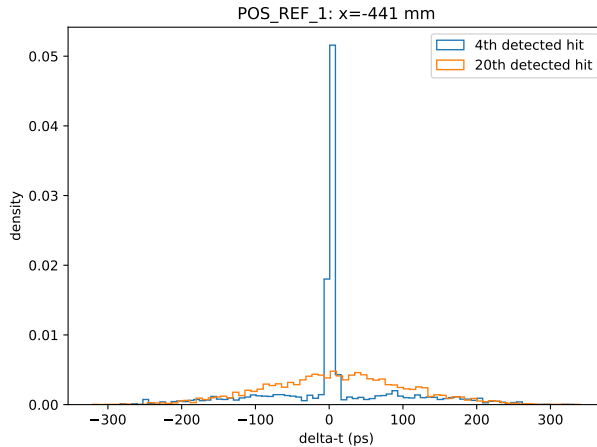
Slope and effective velocity



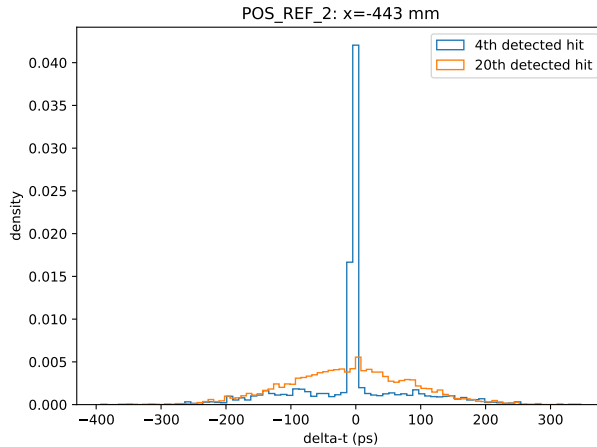
Non-Gaussian tails



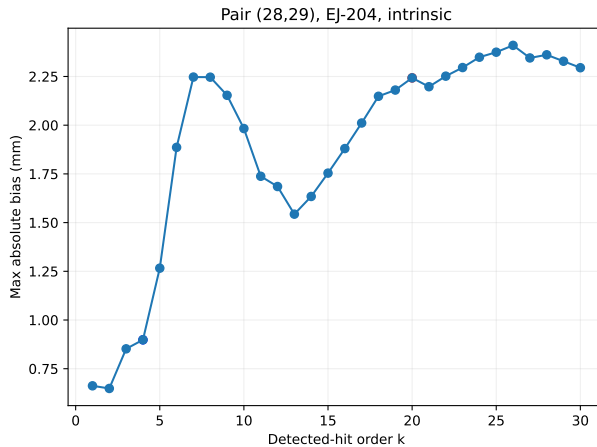
REF1 comparison



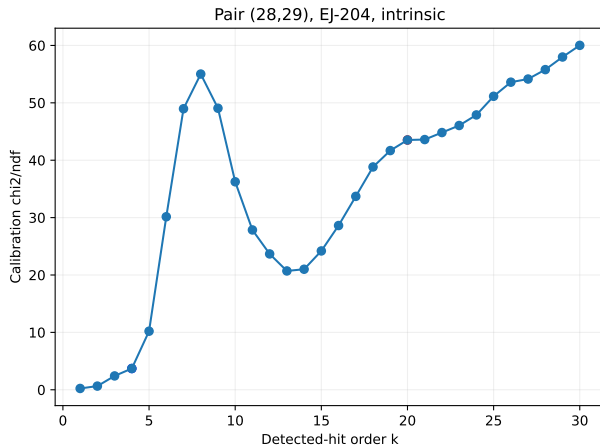
REF2 comparison



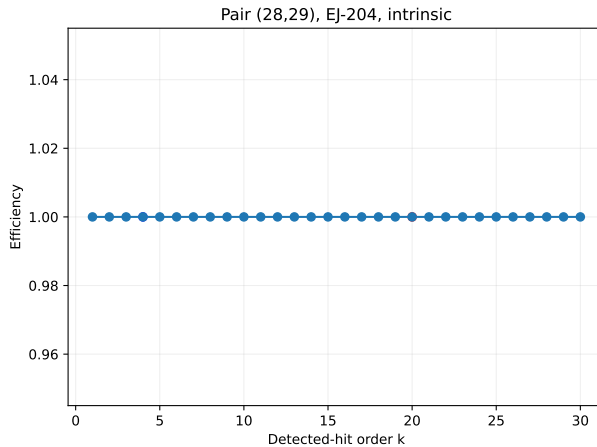
LOO position resolution



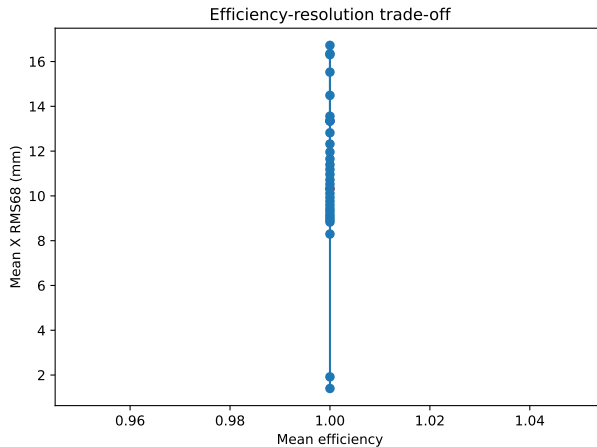
Bias and tails



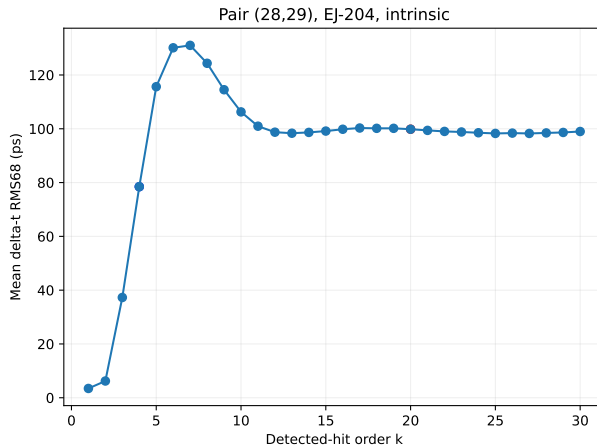
Threshold efficiency



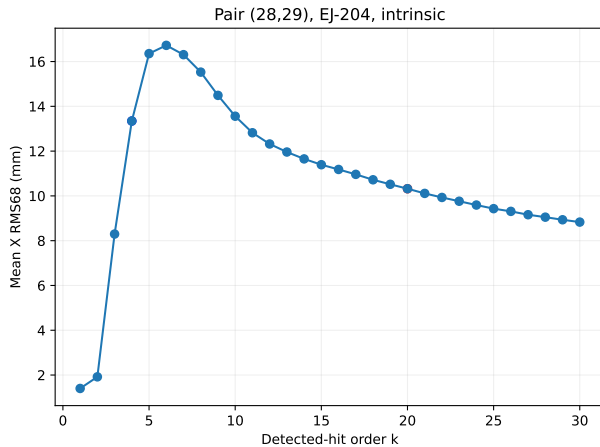
Threshold temporal width



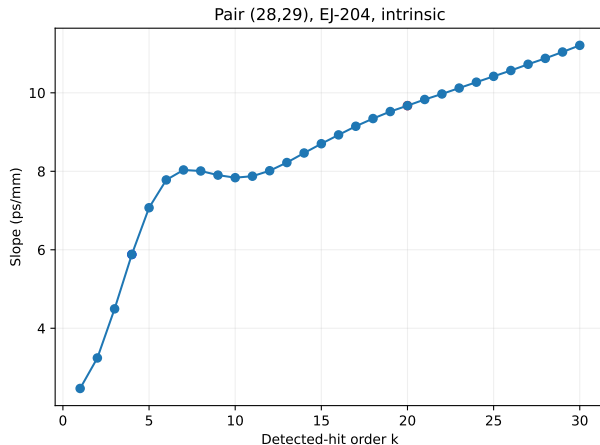
Threshold position resolution



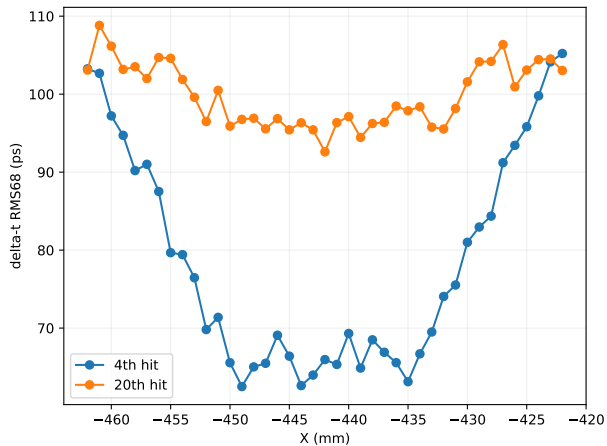
Pareto trade-off



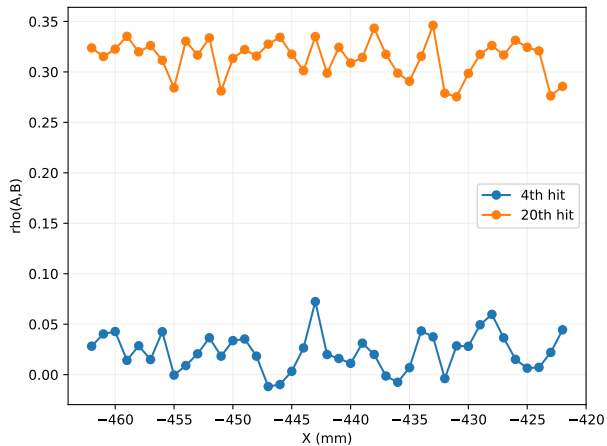
Does 20th improve timing?



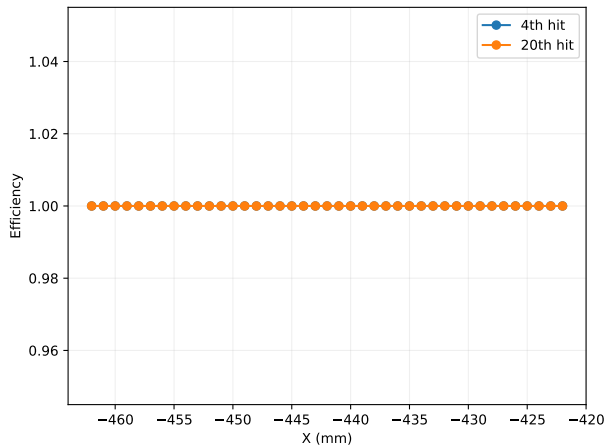
Common-time estimator



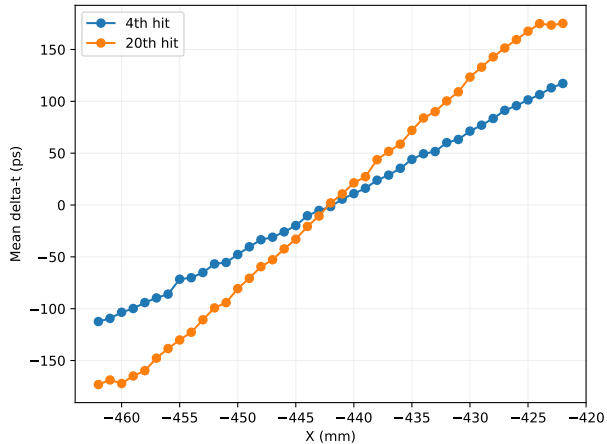
Window dip geometry and light



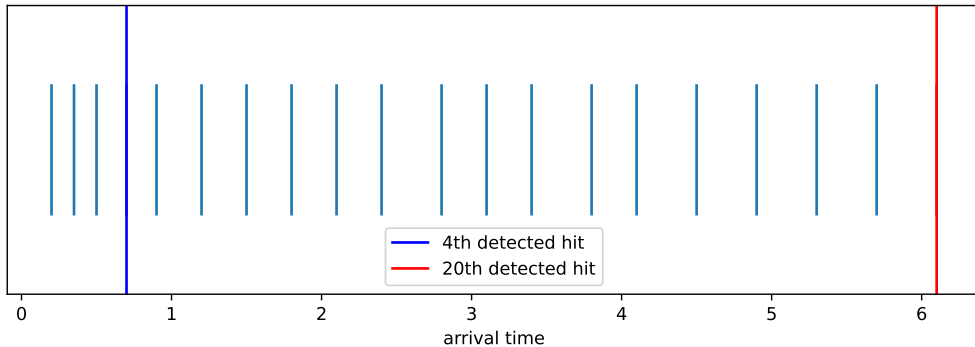
Window dip timing

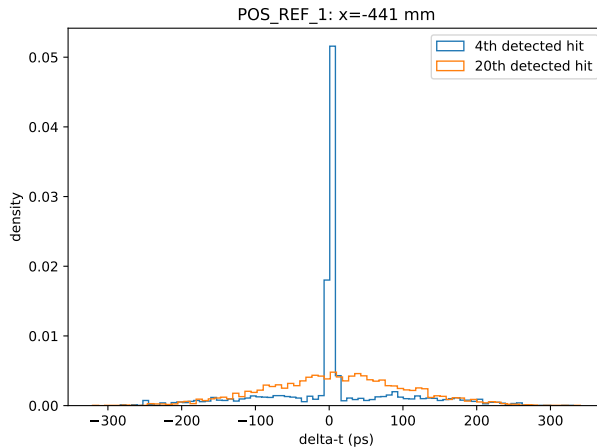


Global EndTop context

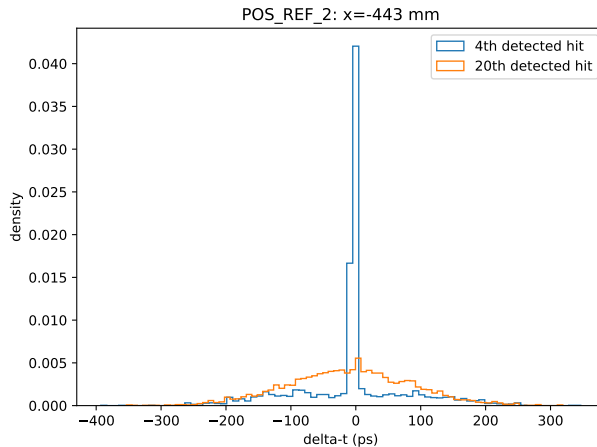


Gap and edge effects

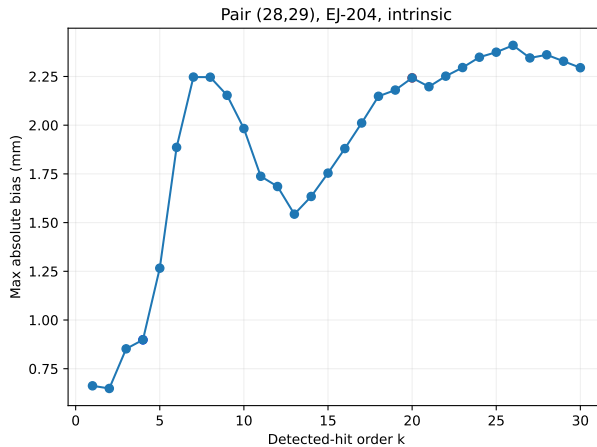




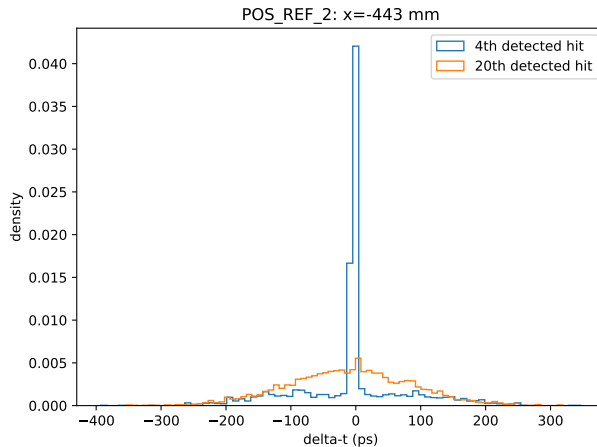
Comparison with Lv et al.



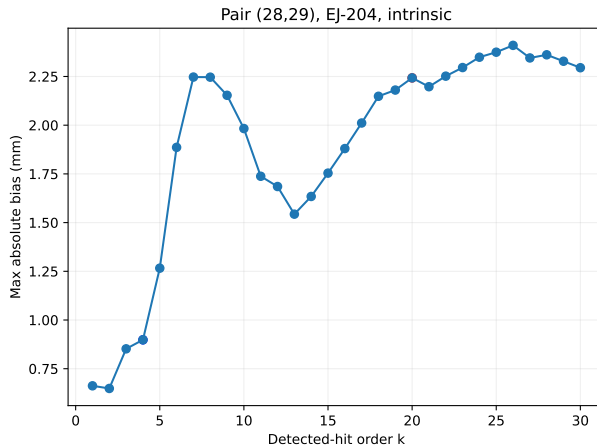
Timing budget



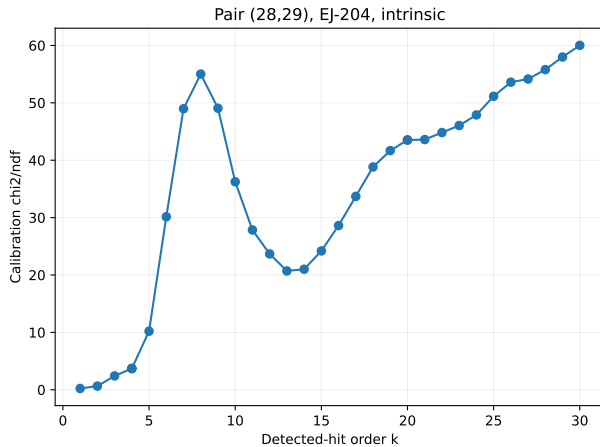
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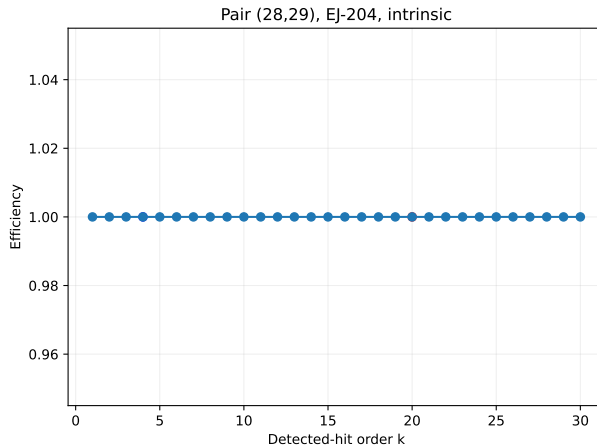
Backup: Schema



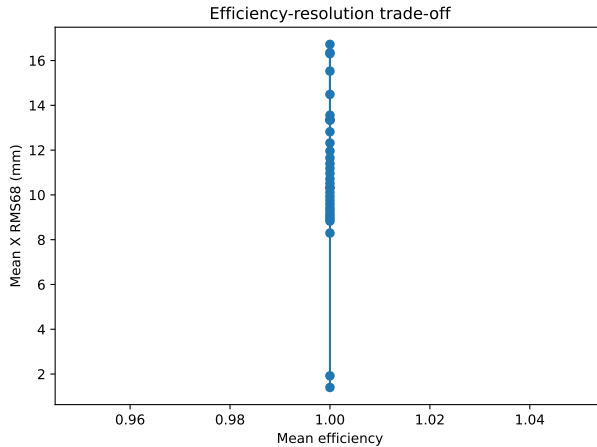
Backup: Builder tests



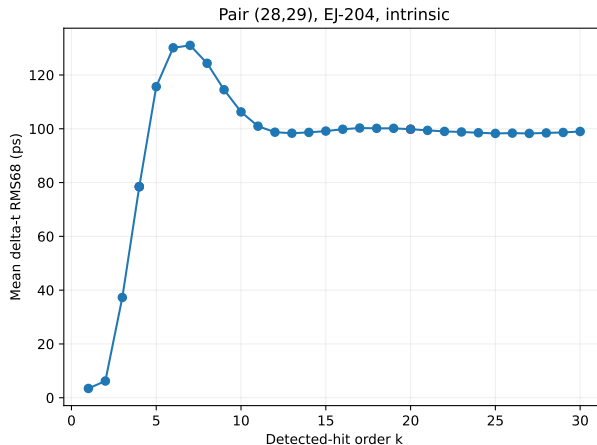
Backup: Fit QA



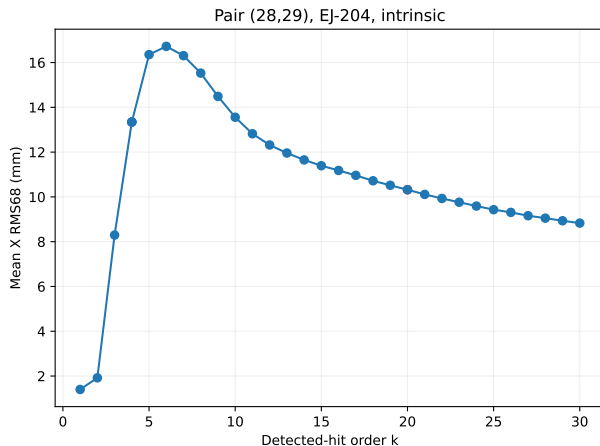
Backup: Threshold table



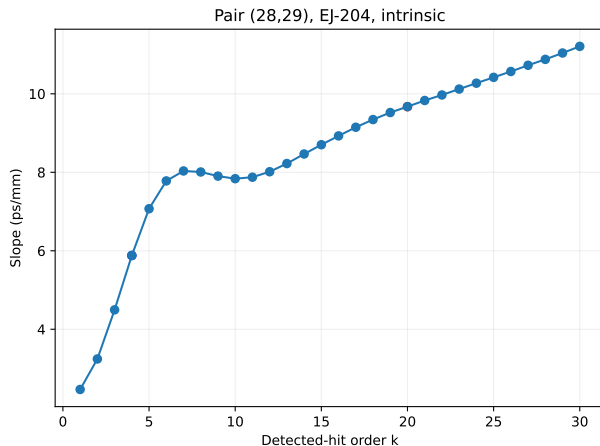
Backup: Calibration covariance



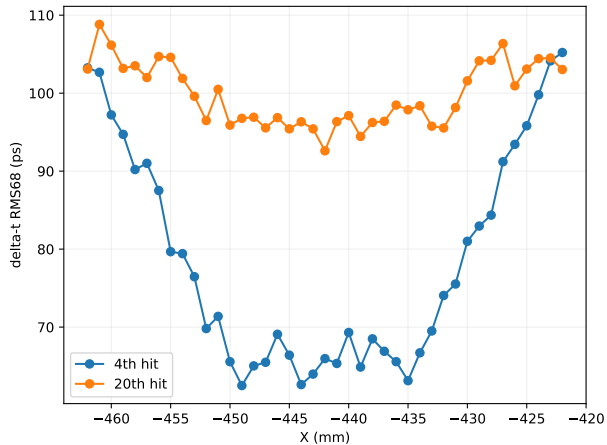
Backup: LOO details



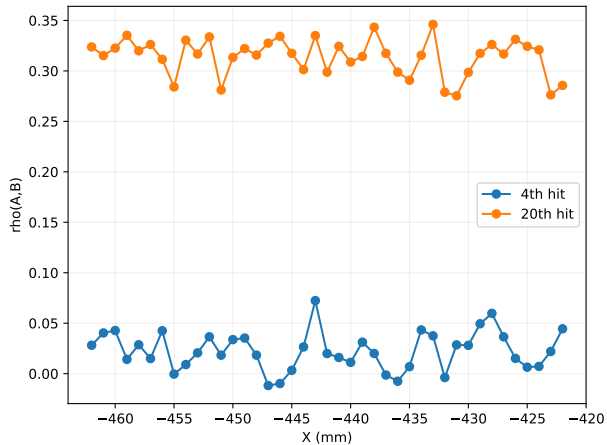
Backup: Tail definitions



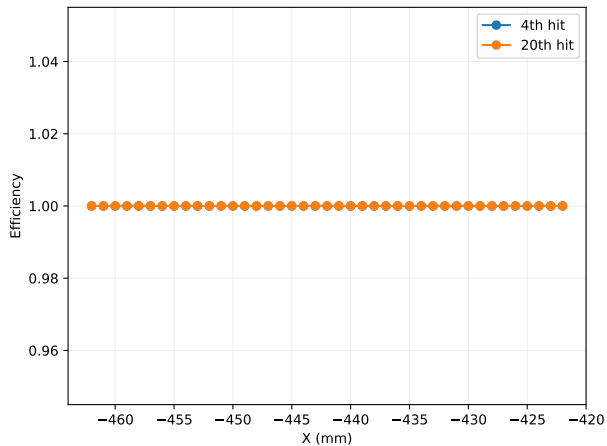
Backup: Window metadata



Backup: Global methods



Backup: Paper comparison



Backup: Reproducibility

